

SECTION AAnswer **all** questions.

1. A hydrocarbon is either cyclopentane or pent-1-ene.

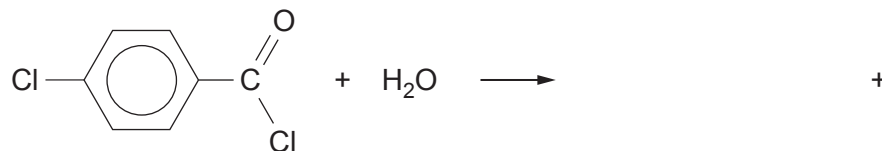
Use the **Data Booklet** to choose a characteristic infrared absorption that will positively identify the compound as pent-1-ene. [1]

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2. Give the **name** of a compound of formula C_4H_8 that will decolourise acidified potassium manganate(VII). [1]

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3. Complete the equation for the reaction shown below. [1]



4. Give the structure of a compound of molecular formula $C_4H_8Br_2$ whose low resolution 1H NMR spectrum consists of three peaks. [1]

